Independent Kummer repair closure review

Ten main declarations, nine producer controls and five independent controls passed fresh checking. All 24 named axiom readbacks used standard Lean axioms. Published source and proof bytes matched the reviewed commit. The prospective review plan predates this review.

Derived correspondence. Native row ordering and the left-associated four-term fold match the field definition. Sixteen matrix equations $M^2 = cI$, with $c \neq 0$, imply preservation and reflection of nonzero vectors and equivalence of the comparator for nonzero inputs. The corrected C2/C1 basis change is conditional on its explicit column equations.

Independent controls. A singular projection collapses two distinct axes; it cannot satisfy the nonzero scalar-square premise. Fourth-column and transpose controls check native orientation. Existing controls cover zero scalar, zero input and incorrect degree-one basis reuse.

Scope. Concrete runtime matrix bindings, complete canonical word-fold correspondence, chart interpretation, xadd geometry and branch existence remain separate. Historical action counts were not rerun. No scientific execution or cost measurement occurred.

Custody and failures. Producer and reviewer differ; historical familiarity is disclosed. Source-first plan, publication binding, fresh replay, controls and readbacks are recorded. Cached dependencies remain trusted. One reviewer simplifier failure was corrected and retained. Individual module exits establish the final result; the first wrapper erroneously returned zero on a failed module. All six final module exits are zero. Source fold $\longrightarrow$ matrix

composition $\longrightarrow$ nonzero and comparator closure.